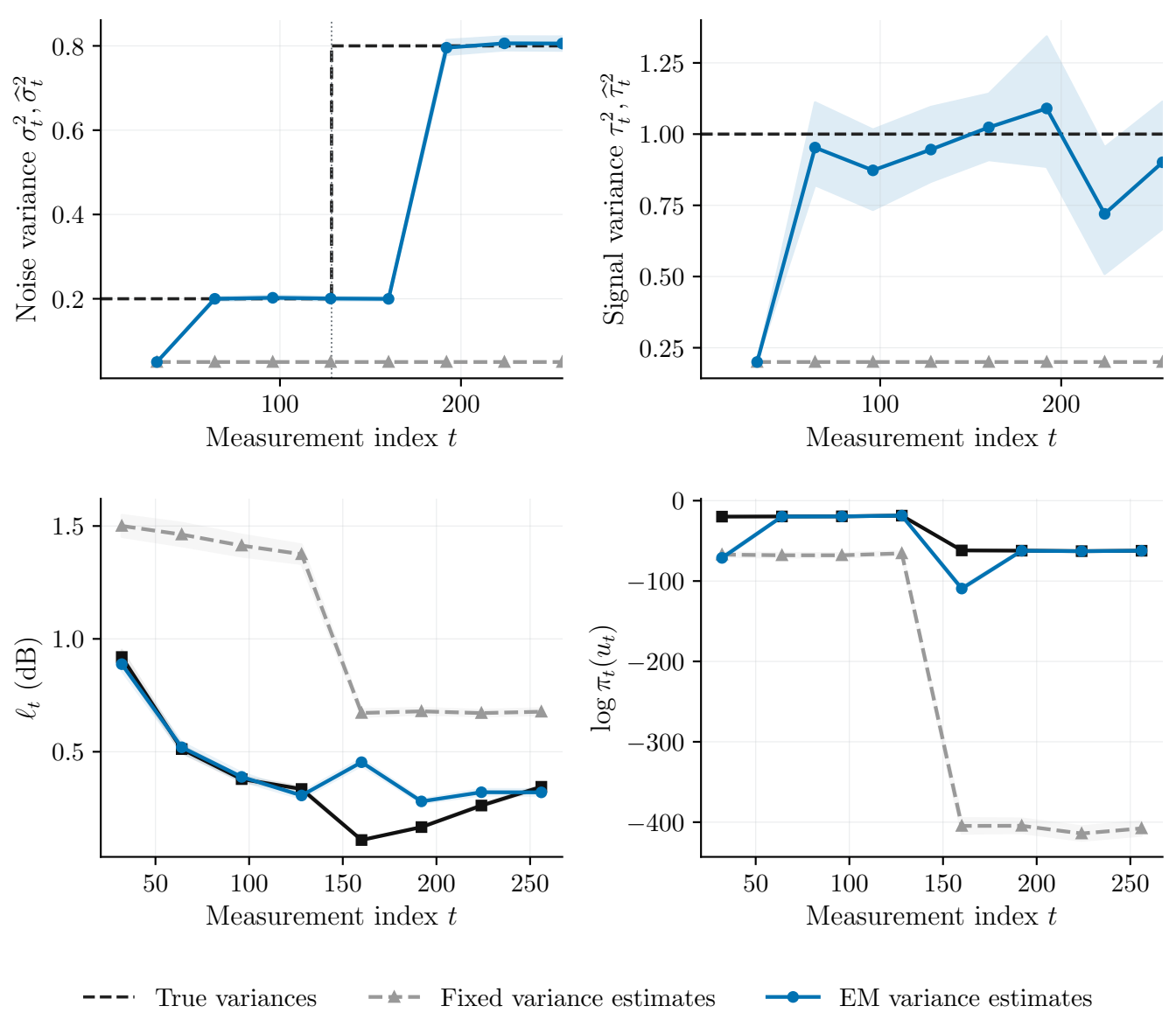
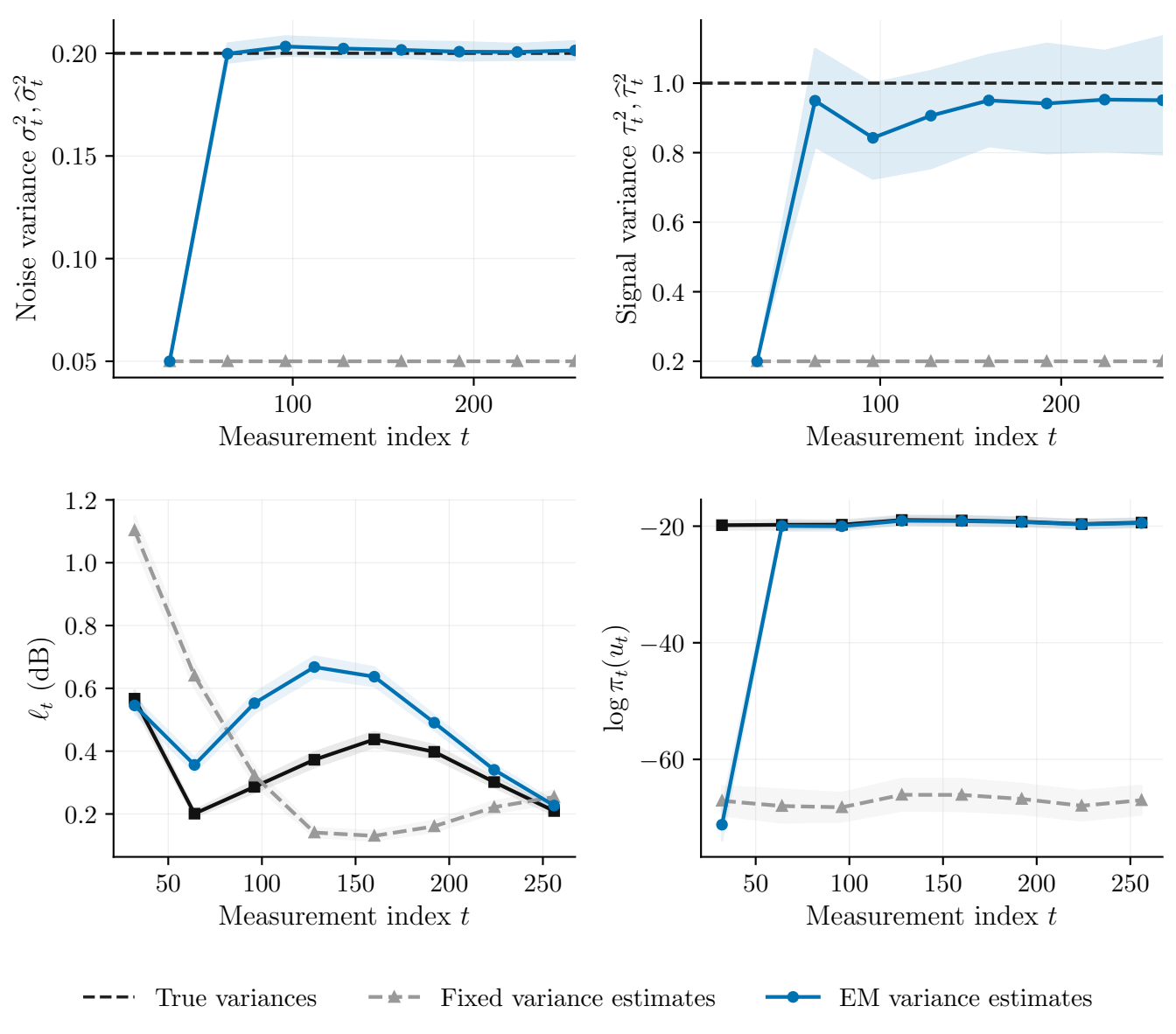
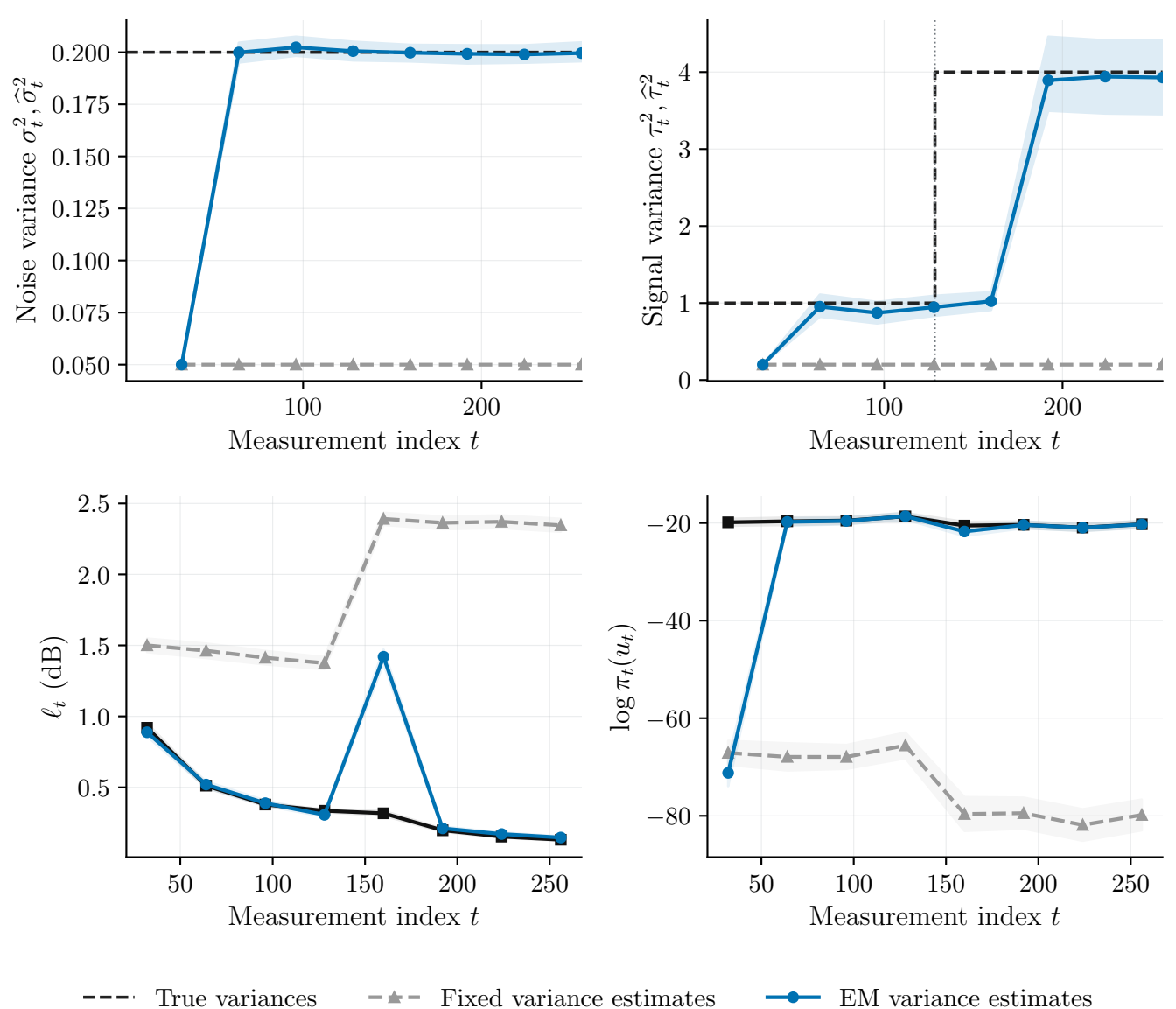

Noise variance $\sigma_t^2, \widehat{\sigma}_t^2$
0.8
0.6
0.4
0.2
100
200
Measurement index t
Signal variance $\tau_t^2, \widehat{\tau}_t^2$
1.25
1.00
0.75
0.50
0.25
100
200
Measurement index t
ℓ_t (dB)
1.5
1.0
0.5
50
100
150
200
250
Measurement index t
$\log \pi_t(u_t)$
0
−100
−200
−300
−400
50
100
150
200
250
Measurement index t
True variances
Fixed variance estimates
EM variance estimates


Noise variance $\sigma_t^2, \widehat{\sigma}_t^2$
0.20
0.15
0.10
0.05
100
200
Measurement index t
Signal variance $\tau_t^2, \widehat{\tau}_t^2$
1.0
0.8
0.6
0.4
0.2
100
200
Measurement index t
ℓ_t (dB)
1.2
1.0
0.8
0.6
0.4
0.2
50
100
150
200
250
Measurement index t
$\log \pi_t(u_t)$
−20
−40
−60
50
100
150
200
250
Measurement index t
True variances
Fixed variance estimates
EM variance estimates


Noise variance $\sigma_t^2, \widehat{\sigma}_t^2$
0.200
0.175
0.150
0.125
0.100
0.075
0.050
100
200
Measurement index t
Signal variance $\tau_t^2, \widehat{\tau}_t^2$
4
3
2
1
0
100
200
Measurement index t
ℓ_t (dB)
2.5
2.0
1.5
1.0
0.5
50
100
150
200
250
Measurement index t
$\log \pi_t(u_t)$
−20
−40
−60
−80
50
100
150
200
250
Measurement index t
True variances
Fixed variance estimates
EM variance estimates